

The Arithmetic of Suspicion

A statistical look at what the Thirlwall Inquiry's recommendations could set in motion

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In brief

This article explores the results of a simulated statistical interpretation of the Thirlwall recommendations. It is not a forecast of how trusts will actually implement them.

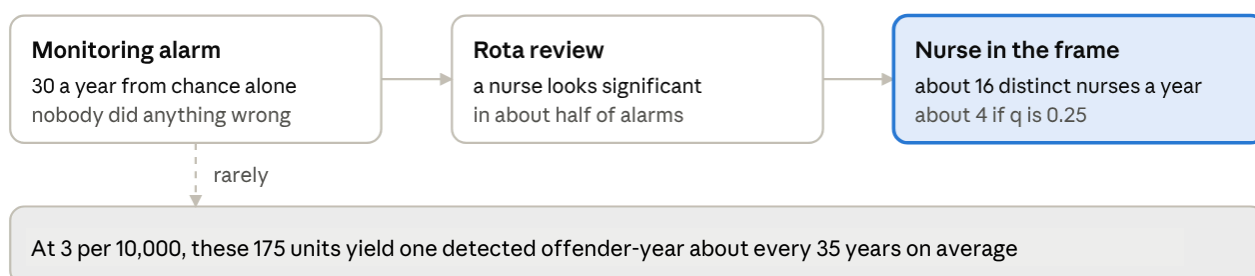
At assumed rates of 1 to 3 offender-years per 10,000 unit-years, the model yields **about 250 to 3,500 false alarmed unit-years per detected offender-year**. Tightening the chart improves that ratio only by sacrificing detection: it approaches 70 to 1 at a setting where, with 3 per 10,000 and 175 units, the system records one **detected offender-year only about once every 4,000 calendar years on average**.

The problem may worsen if a mortality signal is followed by a search of the rota. Someone will always be the nurse most often present, and under a naive significance test, that nurse looks statistically suspicious in about half of chance alarms. Yet when a real offender is present, the same rota exercise can fail to identify her.

One concrete example highlights this. Under a rolling monthly rule used below, chance alone produces about 30 alarm episodes a year. If every alarm prompts a rota search, about 16 distinct nurses a year acquire a nominally significant rota record.

Chance alone could put about 16 nurses a year in the frame

Rolling monthly check (rule C), England and Wales, if every alarm leads to a rota search ($q = 1$)



simulation of 175 units · rule C · base rate 3 in 10,000 unit-years

The alternative to weak monitoring is not broader reporting but stronger investigation. A mortality signal should bring skilled investigators into the unit to examine staffing, workload, infection, equipment, clinical practice and case mix. Only evidence uncovered by that investigation should move the question from “what happened here?” to “did somebody do this?”

This article works through the arithmetic behind this distinction: how often monitoring alarms by chance, what happens when those alarms are matched retrospectively to staff, how much evidential weight the resulting pattern actually carries, and where a statistical safeguard needs to sit if surveillance is not to manufacture suspicion faster than it detects wrongdoing.

The map

Each section answers one question. The short answers come first; the sections then show why.

Section	The question	The short answer
How the recommendations fit together	Where could a chart turn into an accusation?	Where monitoring (Recommendation 6) meets the suspicion protocol (Recommendation 9). The report sets no statistical safeguard there. It calls for consideration of a panel (Recommendation 10) once a person is already in view.
Step one: chance alarms	How often does monitoring raise an alarm when nothing is wrong?	3 to 34 times a year across England and Wales. Even an idealised, perfectly risk-adjusted chart raises 4 to 18.
Step two: the rota	After an alarm, will a nurse look guilty by chance?	About half the time under a naive test; about 2% under a test that allows for someone always coming top.
Step three: the odds	What is an alarm worth as evidence?	Very little: hundreds to thousands of false flags per detected offender-year. Tightening the chart helps only by catching almost no one, and the rota review often misses a real offender too.
The cost side	What does acting on false alarms cost?	Nurses removed from wards already short of staff, and cameras that would almost never record an offender.

Section	The question	The short answer
What would work better	What should an alarm trigger instead?	A review of care, with thresholds set nationally, independent experts before any suspicion forms, and the rota examined blind.
What it adds up to	What does the argument add up to?	A signal is a summons, not a verdict.
Assumptions and limits	What does the analysis not claim?	It does not forecast how managers will behave; that is left as an explicit dial. Rotas are simplified, and base rates are scenarios, not estimates.

How the recommendations fit together

Three of the recommendations, read together, risk working like a smoke alarm wired to the sprinklers. The report does provide an independent-expert safeguard once concerns about an individual have emerged. What it does not specify is a statistical safeguard governing how an outcome signal becomes a concern about a person in the first place. If that junction is left unguarded, the sprinklers will soak the building whenever someone makes toast.

The parts are these:

1. **The detector.** Recommendation 6 asks every trust, by 31 March 2027, for Board-level monitoring of all deaths of children and babies, with a predetermined route for escalating concerning data trends ([Thirlwall Inquiry, Recommendations](#)). It does not say what counts as a concerning trend. The report's own chapter on data is clear that a signal cannot explain itself: as Sir David Spiegelhalter told the inquiry, a monitoring system "cannot say why something has happened" ([Chapter 35](#)).
2. **The trigger.** Recommendation 9 asks for a national protocol on the steps managers must take when concerns or suspicions are raised that a healthcare professional may have deliberately harmed a patient. Whether the manager believes the concern is irrelevant; it must be acted on unless it is obviously irrational or malicious. In Thirlwall's own words, suspicion is enough.
3. **The sprinkler.** Moving the staff member is likely to be the first step, and the report describes the removal as a neutral act ([Irish News](#)).

4. The safeguard. Recommendation 10 asks DHSC to consider setting up a panel for swift technical investigations when concerns involve an individual and harm. It would act after this junction, once a person is in view. The examples given do not name a statistician, though other experts could be included.

The question for a statistician is simple. If a statistical alert is allowed to feed a search of the rota, how often will it produce a "concern" when nothing is wrong, and who gets soaked when it does? The rest of this note answers that in three steps: how often chance alone produces an alarm, who will be in the frame when it does, and what an alarm is actually worth as evidence. How often an alarm does lead to a rota search is a matter of behaviour, not arithmetic, so it is left as a dial.

Step one: how often the alarm sounds by chance

Chance alone will set off somewhere between 3 and 34 alarms a year across England and Wales, depending on the rule chosen. Nobody need have done anything wrong.

A unit's death count behaves like a coin tossed a handful of times: with small numbers, the result wanders far from the average. A local unit that expects four deaths a year will see eight in some years with no change in care at all. Monitoring has to tell that wandering apart from a real shift, and with numbers this small it cannot do so cleanly.

The model. I simulated about 175 neonatal units, roughly the UK's "just under 200" ([Mother&Baby, citing Bliss](#)) less Scotland's 15 ([Scottish Perinatal Network](#)) and Northern Ireland's few. They come in three kinds:

- 45 intensive care units, averaging 20 deaths a year
- 85 local units, averaging 4
- 45 special care units, averaging 1

That gives about 1,300 deaths a year on units, roughly three-quarters of the 1,766 neonatal deaths in England and Wales in 2023 ([Tommy's, citing ONS](#)); the rest happen elsewhere. Each unit's sickness mix varies from year to year (a 25% coefficient of variation), and rules A-D compare a unit with its own average over the three years before the window being checked. I ran ten years of monitoring, 2,000 times over for each kind of unit.

The report does not say what counts as a "concerning trend", so I tested four simple rules a trust might adopt, plus a gold-standard benchmark:

Rule	What triggers it	Unit-years with an alarm, per year	Distinct alarm episodes, per year
A	Year's deaths above the usual 2-standard-deviation line, checked once a year	11	11
B	The stricter 3-standard-deviation line, checked once a year	3	3
C	2-standard-deviation line on a rolling 12 months, checked every month	32	30
D	A deliberately simple heuristic: rolling 12 months shows deaths have doubled (at least 4), checked monthly	34	34
E10	Idealised benchmark: a perfectly risk-adjusted CUSUM chart tuned to a doubling of deaths, set to 1 false alarm per 10 unit-years	17	18
E50	The same chart set to 1 false alarm per 50 unit-years	4	4

Rule D falls hardest on the local units: about one unit-year in three trips it. Rule C shows the price of looking every month; each extra look is another chance for noise to cross the line. Counting distinct episodes instead of unit-years makes almost no difference: a run of alarms rarely straddles two years.

The E rules answer the obvious objection that real monitoring would be better than A to D. The report itself describes risk-adjusted sequential charts of the kind PICANet uses for paediatric intensive care ([Chapter 35](#)). Rule E is an idealised version: it knows each month's true expected deaths, so variation in case mix causes it no false alarms at all. Its false-alarm rate is not a finding but a setting, fixed in advance at two operating points. It can be set stricter still, at the cost of lower detection; step three shows what that trade-off buys. It is tuned to a doubling of deaths, so it is not the optimal detector for every alternative in every kind of unit, but it is a demanding benchmark.

These counts are not an artefact of my assumptions. Give every unit perfect knowledge of its true death rate and remove all year-to-year variation in case mix, and rules C and D still raise 13 and 16 chance alarms a year.

The Shipman Inquiry met the same problem. When Imperial College ran a monitoring chart over 1,009 GPs, 33 crossed the threshold; Shipman was one (politics.co.uk). The researchers judged that the others were almost certainly statistical or medical false alarms ([Plus magazine](#)). A later study found that such charts only reliably caught a doctor causing 10 excess deaths a year, while raising false alarms for more than 15% of practices ([Guthrie et al., BJGP 2008](#)).

Step two: who ends up in the frame

When an alarm sounds, someone will look at the rota, and in about half of cases one nurse will look statistically significant by chance alone.

This is the birthday problem in uniform. Put 23 people in a room and two will probably share a birthday, not because birthdays cluster but because there are many pairs to compare. Put 40 nurses on a rota and ask who was on duty for the deaths, and one of the 40 will almost always have been there more than the rest.

The model. Every chance alarm from step one is passed straight to a review of the deaths in the window that set it off. Staff numbers follow unit size: 100 nurses in an intensive care unit, 40 in a local unit, 20 in a special care unit. Half work full time, three 12-hour shifts out of 14 a week, so they are on duty about 21% of the time. A third work part time (about 13%). The rest pick up extra shifts (about 27%). Each death falls on a random shift, with nobody causing it.

I then tested the record of the most-present nurse in three ways (the technical notes call them the average-exposure, own-exposure and maximum-adjusted tests):

1. **The careless test.** Compare her attendance with the average nurse's share of shifts. This is what a bare "she was on duty for X of Y" invites.
2. **The naive test.** Compare her attendance with her own share of shifts, but as if she had been named before anyone looked.
3. **The correct test.** Ask how often the top nurse on the whole rota would do at least this well by chance, given every nurse's actual share of shifts. Under the simulated null, where each nurse attends independently at her own rate, this can be calculated exactly. A real rota, with fixed staffing, teams and night shifts, would need a test built on the rota itself.

Unit type (rule C alarms)	Deaths reviewed (median)	Top nurse attended (median)	Looks significant: careless test	Naive test	Correct test (under this model)
Intensive care	28	12	100%	69%	3.1%
Local	8	4	68%	51%	1.8%

Unit type (rule C alarms)	Deaths reviewed (median)	Top nurse attended (median)	Looks significant: careless test	Naive test	Correct test (under this model)
Special care	3	2	26%	26%	1.4%
All alarms, weighted by unit numbers	—	—	76%	55%	2.3%

The correct test behaves as it should when the deaths are random: it flags only about 2%, below the nominal 5% because attendance counts are whole numbers. The naive test finds a "significant" nurse in more than half of all alarms, because she was not named in advance. She was picked *because* she was the highest of 40 or 100. Testing her as if she were the only candidate is the Texas sharpshooter's method: fire at the barn, then paint the target round the tightest cluster of holes.

The careless test does worse still, because the nurse who tops the rota is usually one who works more shifts than average. It is a second, separate error, but a common one: a rota chart with no denominator commits it by default. The other rules give much the same picture: across all six, the naive test flags a nurse in 49% to 60% of alarms, and the correct test in about 3% or fewer.

The model is generous to the process. Real reviews can make it worse in three ways that the Royal Statistical Society report on these cases describes ([Green, Gill et al. 2022](#)):

- **Loose events.** Collapses, desaturations and "unexpected deteriorations" join the deaths, multiplying the events to match against the rota.
- **Flexible windows.** The period under review is chosen after the cluster is seen.
- **Selective counting.** Once a name is in the air, events she missed tend to be explained away, and events she attended tend to be kept.

Step three: what an alarm is worth

At illustrative rates of 1 and 3 offender-years per 10,000 unit-years, hundreds to thousands of unit-years are falsely flagged per detected offender-year. Even at 30 per 10,000, the ratios span about 25 to 120 across these monitoring rules.

A smoke alarm in a kitchen is trusted because fires are common there and toast is easy to recognise. A neonatal unit is the reverse: the fire is vanishingly rare and the toast looks exactly like smoke. When the thing you are looking for is that rare, even a good detector mostly finds noise.

The base rate. Killing by staff in neonatal care is extremely rare, but its true rate is unknowable from convictions. The UK has seen one neonatal case in recent decades, Lucy Letby's, and her convictions are disputed; Beverley Allitt worked on a children's ward, not

a neonatal unit. Rather than defend a single figure, I use four scenarios: 1, 3, 10 and 30 offender-years per 10,000 unit-years. These are scenarios, not prevalence estimates. The higher two test deliberately pessimistic assumptions.

Detection. I then put an offender on the rota of each kind of unit. She works a full-time pattern, causes an expected four (or seven) extra deaths over twelve months, and is present at every death she causes. I asked how often each rule caught the unit within that year, and how often the rota review then named her:

Rule	Unit caught: intensive care, +4 expected deaths	Local, +4	Special care, +4	Local, +7 (Letby-scale)	Rota review names her: local, +4
A, annual 2-sigma	24%	40%	66%	71%	59%
B, annual 3-sigma	10%	17%	38%	44%	61%
C, rolling, monthly 2-sigma	40%	55%	73%	79%	29%
D, doubling heuristic	11%	67%	77%	86%	23%
E10, idealised, 1 in 10	25%	55%	83%	84%	40%
E50, idealised, 1 in 50	8%	23%	53%	60%	57%

Combining detection and chance alarms across all 175 units gives the number of falsely flagged unit-years for each detected offender-year ("detection" below means the unit was flagged, not that the offender was identified):

Rule	Base rate 1 in 10,000	3 in 10,000	10 in 10,000	30 in 10,000
A	1,510	500	151	50
B	850	284	85	28
C	3,220	1,070	322	107
D	3,530	1,180	352	117
E10	1,790	600	179	60
E50	740	248	74	25

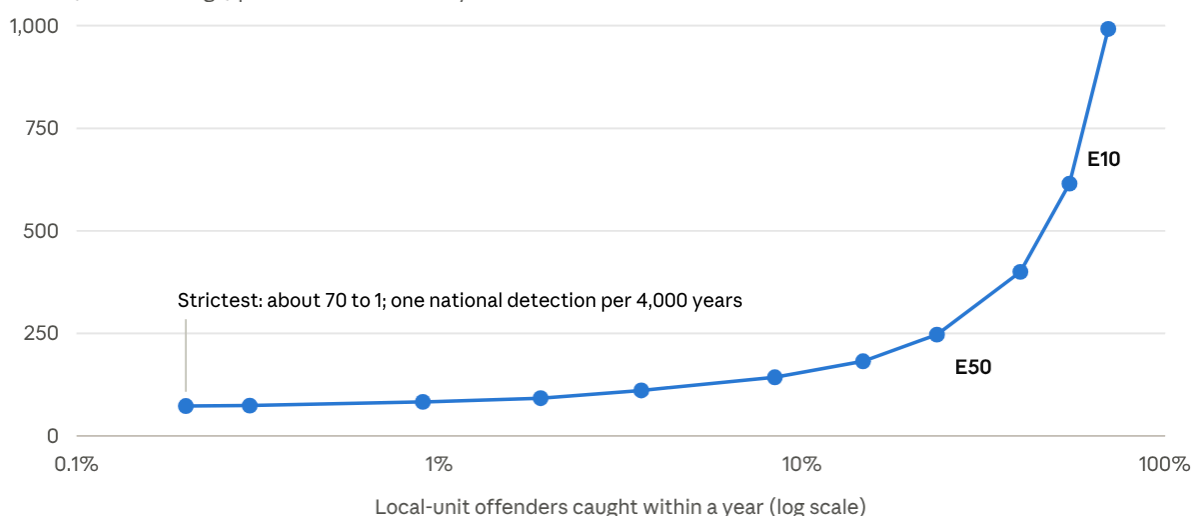
None of the six rules achieves both high sensitivity to four extra deaths and a useful predictive value for deliberate harm. The strict rules (B and E50) keep false alarms down but miss most offenders. The sensitive rules catch more, at the cost of a thousand or so false alarms per detection in the 1 to 3 per 10,000 scenarios. At 3 per 10,000 across 175 units, the model produces one detected offender-year every 34 to 35 calendar years under C and D, or every 71 to 93 years under B and E50.

The idealised chart shows a trade-off, not a fixed floor. A tighter threshold lowers false alarms and improves the ratio, but detection also falls. The ratio nears 100 to 1 only when the chart catches about 3% of local-unit offender-years within a year. At the strictest tested setting it is roughly 70 to 1; at 3 per 10,000, the 175-unit system records one detected offender-year about every 4,000 years. These tail results are approximate because calibration at that setting rests on about 50 simulated crossings.

Tightening the chart improves the odds only by catching almost no one

Idealised risk-adjusted CUSUM at eleven settings, England and Wales, base rate 3 in 10,000 unit-years

False alarms (chart crossings) per detected offender-year



thirlwall_statistical_model.py, output block 9 · idealised CUSUM tuned to a doubling, calibrated on fixed null paths · +4 expected deaths · detection shown for local units, ratio for all 175 units

E10 and E50 are the two settings used in the tables. False alarms can always be driven lower by tightening the monitor. But at these base rates, getting anywhere near 100 false alarms per detection requires a monitor so strict that it detects almost nothing.

Four further results matter. First, the rota review is weak in both directions. When there is an offender and the alarm fires, the review names her in only 23% to 61% of local-unit cases, and in under 15% of intensive care cases, because her deaths are diluted among the unit's natural ones. Second, spreading the same national base rate in proportion to staff numbers, so that big units carry more of the risk, makes every ratio worse, not better,

since offenders in large units are harder to see. Third, giving the offender exactly four deaths rather than an expected four moves the ratios by about 5% to 20% either way and reverses nothing.

Fourth, detection is defined generously: any alarm in a year when an offender is present counts, whether or not her deaths caused it. Running the same rules on the same kinds of unit with no offender shows how much of that "detection" is background. Under rule C, 18% of local units and 27% of intensive care units would be flagged in that year anyway, against 55% and 40% with an offender present. Under rule D in local units the figures are 33% and 67%: half of what counts as detection would have happened regardless. The idealised chart does better here, with 7% background against 55%. The predictive values above are therefore, if anything, flattering.

Putting the steps together. Call q the share of alarms followed by a rota search. Each unit keeps the same staff for the whole ten years, so the model can count distinct nurses as well as flagging episodes; the two barely differ, because repeat flags of the same nurse are rare. Nurses each year whose record passes the naive test:

Rule	$q = 1$	$q = 0.5$	$q = 0.25$	$q = 0.1$
A	6.2	3.1	1.6	0.6
B	1.8	0.9	0.4	0.2
C	16.4	8.2	4.1	1.6
D	16.2	8.1	4.0	1.6
E10	8.6	4.3	2.2	0.9
E50	1.8	0.9	0.5	0.2

The report does not set q at 1. Its data chapter treats a signal as a prompt for review, and Recommendation 9 is triggered by concerns about a person, not by a chart. Readers can choose their own q . The one documented case is Chester itself, where the rota was examined because deaths had risen.

This is where a low threshold for suspicion becomes statistically consequential. With the offender on the rota, the model can also say how likely it is that a flagged nurse is in fact the offender. At 3 in 10,000, it is between 0.06% and 0.5% depending on the rule. In the most pessimistic case tested, the strictest risk-adjusted chart at 30 in 10,000, it reaches 4.7%. Recommendation 9 treats any good-faith concern as enough to act on, whatever the manager believes, and the report calls the action neutral.

The cost side

The main cost falls on staffing, which is already the weakest point in neonatal care. The cameras cost money but are unlikely ever to catch anyone.

Removals. Suppose each nurse put in the frame is removed while an investigation runs. Dr Martin Pitman's own investigation was promised at six weeks and took eighteen months. At that pace, rule C with every alarm followed up ($q = 1$) would keep about 25 nurses off neonatal wards at any one time; at $q = 0.25$, about 6. Those figures cover data-driven alarms only. Concerns raised by colleagues, which the protocol also covers, cannot be modelled and could easily outnumber them.

Neonatal units can ill afford the loss. In 2022 only 71.1% of nurse shifts in England, Wales and Scotland were staffed to recommended levels, and the figure had fallen for two years running ([Infant journal, on the NNAP report](#)). Staffing matters to survival: a 2007 study found that bringing specialist nurse ratios up to one-to-one was associated with a 48% fall in risk-adjusted mortality among the smallest babies ([Bliss evidence to Parliament](#)). A safeguard that removes nurses on chance alarms works against the very thing it is meant to protect. The effect on recruitment, of nurses knowing that a bad month can end a career, is real but cannot be measured in advance.

Cot cameras. The report wants monitors on every cot and incubator so parents can watch remotely ([ITV News](#)). The North West has 469 cots across 21 units, about 22 a unit ([NHS England North West](#)), which implies roughly 3,500 to 4,000 cots in England and Wales. At 3 offender-years per 10,000 unit-years, 175 units over a decade imply an expected 0.5 offender-years, however many cots carry a camera. The 37,000 cot-years of footage measure the scheme's scale and cost, not its chance of a catch. Its case is reassurance and deterrence, not detection, and should be argued on those grounds.

Insulin CCTV. Cameras on insulin fridges, with recordings kept at least 28 days ([Pharmaceutical Journal](#)), help only if suspicion arises within that window and the insulin came from that fridge. A central dispute in the Letby insulin cases concerns something else: an immunoassay result that was never confirmed by a more specific test. Routine mass-spectrometry confirmation of unexpected insulin results would cost less, and would protect against wrongful accusation as well as against poisoning.

What would work better

Monitoring is worth having if an alarm leads to a review of care, not a search for a culprit. The fix is to break the wire between detector and sprinkler.

1. **Set the rule nationally, with a stated false-alarm rate.** Statisticians, not trust boards, should choose the chart and its threshold, adjusted for each unit's case mix. Everyone then knows in advance how many alarms chance will produce.

2. **Treat an alarm as a question about care.** The first response should look at infections, staffing, case mix and equipment, which explain most clusters. Paul Aylin, who built the post-Shipman monitoring system, argued that monitoring should start audit and learning, not recrimination ([Digital Health](#)).
3. **Put independent experts between the alarm and any suspicion.** The Royal Statistical Society recommends that such investigations be supervised by panels independent of both the suspect and the employer ([RSS](#)). It is the kind of panel the National Crime Agency is reported to have advised in the Letby investigation, and which was never convened.
4. **Blind the reviewers.** Classify each death as explained or unexplained before anyone sees the rota. Fix the list of events and the time window first.
5. **Count the whole rota.** If staff presence is examined at all, compare every nurse's record and allow for the fact that someone always comes top.
6. **Separate protection from accusation.** Where a concern is serious, a change of duties agreed with the nurse, without a finding, protects babies without branding anyone. It is closer to the "neutral act" the report describes than suspension is.

What it adds up to

A mortality signal is a summons, not a verdict.

The three steps form a chain. Chance alone sets off several to dozens of alarms a year. Matching the deaths behind an alarm to the rota then produces, in about half of cases, a nurse who looks statistically significant, though every death in the model is random. And even a perfectly risk-adjusted monitor cannot rescue the odds. In the lower-rate scenarios, false flags outnumber detections by hundreds to one or more, and tightening the threshold helps only by catching almost no one. When a real offender is present, the same rota review often fails to name her.

None of the recommendations creates this risk on its own. Monitoring is sound; so is acting on genuine concern; so is expert review. The risk lies at the junction where an outcome signal becomes suspicion of a person, and that is the one point the report leaves without a statistical safeguard. How often trusts cross it is a matter of behaviour, which the analysis leaves open as an explicit dial. What happens when they do is a matter of arithmetic.

A mortality chart can tell you where to look. It cannot tell you whom to blame. The answer to a signal is not stronger suspicion. It is stronger investigation: an independent review of care, with the statistics set in advance and the rota examined blind. What the analysis does not claim is set out below.

Assumptions and limits

The numbers are illustrative, not forecasts. They show orders of magnitude; the reported sensitivity checks show which conclusions persist as assumptions change.

- **Unit mix, death rates and staffing are modelled.** The split of 45 intensive care, 85 local and 45 special care units, their average deaths and their staff numbers (100, 40, 20) are my assumptions, informed by published national totals. Real units may differ and produce different alarm rates.
- **The monitoring rule is unknown.** The report leaves it open. Rules A-D are deliberately simple. Rule E is an idealised CUSUM benchmark with knowledge of the true monthly expectation, an advantage a real monitor would not have.
- **Baselines are recent history.** Rules A-D use the three years before each window; E uses the true monthly expectation. Annual rates vary, but long-run means are fixed.
- **Rotas are simplified.** Each nurse's presence at each death is independent of everyone else's. In reality deaths cluster at night and on busy shifts, and some nurses work more of those. That can make chance coincidences more likely, not less.
- **Behaviour is a dial, not a forecast.** How often an alarm leads to a search for a person (q) is left open and shown at four values.
- **The base rate is a scenario, not an estimate.** Four values from 1 to 30 per 10,000 unit-years are shown; the conclusion holds across all of them, and holds more strongly if the risk is spread in proportion to staff numbers. The offender is modelled as a full-time nurse present at every death she causes; nothing else marks her out.
- **Paediatric wards are left out.** The report's monitoring covers deaths of children as well as babies, which would add further alarms.
- **Detection is generous.** Any alarm in a year when an offender is present counts as detecting her unit, whether or not her deaths caused it. Step three shows how much of it is background.
- **Units of the false-to-true ratio.** The tables compare falsely flagged unit-years with detected offender-years; the threshold sweep counts chart crossings instead. These are distinct measures and should be compared with that difference in mind.
- **Posterior events.** The offender side counts the first alarm in her year, while the background side counts every flagged episode. This asymmetry affects interpretation of the resulting probability.

The simulation underlying the main numerical results is documented in the accompanying [source code](#) and [technical notes](#), with the full model specification in [MODEL.md](#). Readers can rerun it and test different assumptions or monitoring rules. This is one explicit model and interpretation, open to comparison with others. The changes made in each round of review are listed in the accompanying README's Version history.

Sources

- [ITV News: 14 recommendations](#), 15 September 2026
- [IBTimes: deliberate-harm protocol](#)
- [Irish News: "suspicion is enough", "neutral act"](#)
- [Pharmaceutical Journal: insulin storage and CCTV](#)
- [Tommy's: neonatal deaths 2023, citing ONS](#)
- [Mother&Baby, citing Bliss: number of UK units](#)
- [Scottish Perinatal Network: Scottish units](#)
- [NHS England North West: cots per unit](#)
- [Infant journal: NNAP nurse staffing](#)
- [Bliss evidence to the Public Accounts Committee](#)
- [Green, Gill, Mackenzie, Mortera and Thompson, *Healthcare serial killer or coincidence?* \(RSS, 2022\)](#)
- [RSS press release on the report](#)
- [Plus magazine: could maths have caught Shipman?](#)
- [politics.co.uk: Aylin et al. GP monitoring results](#)
- [Digital Health: Aylin on monitoring as audit](#)
- [Guthrie et al., *British Journal of General Practice*, 2008](#)
- [Thirlwall Inquiry final report: Recommendations](#)
- [Thirlwall Inquiry final report: Chapter 35, data and signals](#)
- [Thirlwall Statistical Model: source code and technical notes](#)